

Using Blockchain and Vazka Authentication for the Security of Smart Home Devices

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Abstract—Smart home devices are increasingly used to automate daily tasks and provide enhanced comfort and convenience. However, as these devices become universal, their security risks also increase. These risks are caused by a lack of implementation of secure safety methods embedded within these devices and the nature of the data generated by these devices that could be used for malicious activities. In this paper, we propose an innovative robust approach to improve the security of data generated or collected by smart home devices. Our proposed method combines Blockchain with Vazka Authentication. First, our solution involves creating a decentralized system for storing devices collected or generated data using blockchain. Then, we introduce our new authentication mechanism, called Vazka, to limit unauthorized access to smart home devices data. It consists of dynamically generating a password based on the collected data signature. We analyze a particular case study that consists in applying our proposed solution to a smart home security camera's system. Our proposed solution has the potential to significantly enhance the security of video footage generated and collected by the smart home security camera. Overall, our proposed solution makes it more difficult for malicious actors to gain unauthorized access and/or tamper smart home devices collected data.

Keywords— authentication, blockchain, cyberattack, data integrity, smart home devices, tamper-proof.

I. INTRODUCTION

Our everyday lives have changed as a result of the quick spread of smart home technology, which provides more comfort, convenience, and control over our living spaces [1]. These benefits do, however, come with certain inherent hazards. The risk of security lapses and cyberattacks is growing along with the number of networked devices. Malicious people may take advantage of security flaws in smart home gadgets to obtain access without authorization, compromise user privacy, and even cause bodily harm [2].

For instance, data collected from smart home devices such as smart energy meters can reveal more about a family than just the amount of energy used. For example, energy usage data can reveal when a person is home and when he is away, when he has company and when he may be alone, and more.

On the other hand, the complex cyber threats that are present today cannot be adequately fought back using the security methods used currently in smart home devices such as passwords and two-factor authentication. As a result, stronger security measures are urgently needed to safeguard smart houses and guarantee the safety of their occupants [1], [3].

Advanced security solutions are required since hackers always create new ways to attack vulnerabilities. Furthermore, many smart home systems have a single point of failure due to their centralization, making them vulnerable to widespread assaults [4], [5]. Therefore, it



Fig. 1: Integration of Blockchain Technology in Smart Home Systems

is essential to investigate cutting-edge strategies that might improve the security of smart home devices.

For that purpose, we aim in this study to provide answers to the following research questions:

- How might cutting-edge technologies such as blockchain technology be used to improve smart home device security?
- Is it feasible to develop a novel authentication technique that works with blockchain technology and offers high levels of security for smart home devices?
- What possible advantages and difficulties come with the suggested authentication method?

To answer these questions, in this paper, we propose a new solution for securing data of smart home devices. It consists of combining blockchain technology with a novel robust authentication technique. The suggested solution intends to provide a

The rationale behind our proposed solution is that by utilizing the decentralized and transparent characteristics of blockchain technology, we ensure a reliable and tamper-resistant data storage. On the other hand, generating dynamic content-based passwords makes it nearly impossible to crack them.

Our proposed solution is advisable for implementation under the following specified constraints:

- 1) The smart home device's data to be accessed and secured is stored in rest and not on the fly.
- 2) The smart home device is continuously collecting/generating data (i.e the data is updated and changed regularly).
- 3) The access to the stored data is not on a regular basis (not a frequent access).

Examples of scenarios where our proposed solution can be applied can be as follows:

- Smart cameras indoor or outdoor capturing videos 24/7 and storing these videos on a local server.
- Smart speakers listening continuously and storing audio data on a local server.

- Smart devices continuously storing sensitive data (such as personal details, notes, passwords, and banking information).
- Smart meters continuously collecting and storing sensitive data such as smart energy meters.
- Smart displays with built-in cameras and microphones continuously listening to commands and storing logs about played games, watched videos, and web searches history.

Examples of scenarios where our proposed solution cannot be applied because the device or data does not respect the aforementioned constraints, can be as follows:

- Smart locker where data is generated once and it is not continuously changing.
- Smart light sensor where data is only used on the fly and not stored.

Our proposed solution improves the general security of smart home devices by protecting user privacy and maintaining data integrity. A safer and more secure ecosystem for smart home consumers may be promoted by our research's findings, which can also help the smart home sector build standardized security policies and recommendations.

In the remainder of this paper, we first analyze the existing methods for securing smart home devices data in Section II. Then, in section III, we explain in detail our proposed solution. In Section IV, we consider a case study and demonstrate the effectiveness of our proposed method. Finally, in Section V, we conclude this paper.

II. RELATED WORK

Pecorella et al. conducted a study proposing the implementation of network sentiment analysis to enhance security and privacy in smart homes [6]. The approach involved adjusting the security level of the smart home network based on the user's perceived risk level. They suggested distributing firewalls and intrusion detection systems on the home and internet service provider sides and collaborating efforts to effectively respond to security threats. The study also highlighted the importance of tailoring network security based on user-specific factors and considering spatial and social correlations of threats.

Another study by Park et al. focused on a secure device control scheme using blockchain technology in smart homes [5]. They presented an improved technique for operating smart home devices safely and effectively through a blockchain-based Internet of Things (IoT) system [7]. The study emphasized the need to regulate linked IoT devices in a smart home network and addresses latency issues and prevention of forgery attacks.

Additionally, Kumar et al. proposed a lightweight authentication scheme for smart home IoT devices [8]. They highlighted the challenges of data security and resource limitations in IoT networks and suggested an authentication scheme involving an authentication device and a controller device to operate home appliances [9]. Their scheme aimed to thwart various attacks and ensured user and network device identity confidentiality.

III. PROPOSED SOLUTION

A. Solution Overview

As the use of smart home devices increases, safeguarding the growing amount of personal data stored on servers becomes a critical concern. To address this, we propose a novel Authentication scheme combined with Blockchain technology for data security and integrity.

Figure 2 shows our proposed Blockchain and *Vazka* Authentication Scheme. First, smart home devices collect and/or generate data from the environment such as videos from cameras, data from sensors, etc. and send it for storage.

This collected data is distributed on the network nodes in a decentralized fashion using the consensus and replication process [10] of blockchain technology (Block A).

Then, when the user (home owner for instance) requests to access and read data, the Blockchain's retrieval process [11] is applied.

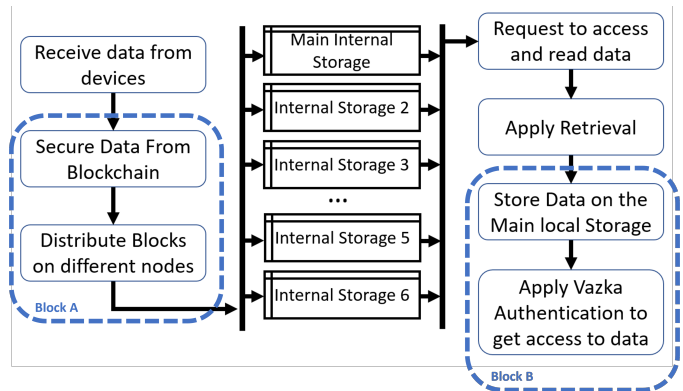


Fig. 2: Proposed Blockchain and Vazka Authentication Scheme

The required blocks are retrieved and stored in the main local storage device. To gain access to the retrieved data, user should authenticate using our proposed authentication method, called *Vazka* authentication (Block B).

B. Implementing Blockchain

The first part of our contribution consists in integrating the blockchain technology within smart home devices. Figure 4 illustrates our proposed blockchain implementation (Block A) in smart home devices:

- Step 1 - Initialization: The smart home device and server generate their respective public and private keys.
- Step 2 - Data Capture and Encryption: The smart home device collects/generates data and encrypts it using its private key. The encrypted data is then sent to the server.
- Step 3 - Verification and Decryption: The server verifies the authenticity of the data by decrypting it using the smart home device's public key. If the data has been tampered or the encryption doesn't match, the server discards it.
- Step 4 - Hashing and adding to Blockchain: Once the server has verified the data, it generates a hash of the data and adds it to the Blockchain along with a timestamp.
- Step 5 - Consensus: Other nodes in the network verify the block's hash and add it to their copy of the blockchain.
- Step 6 - Retrieval: If needed, the data can be retrieved from the Blockchain using its hash and the smart home device's public key.
- Step 7: Periodic maintenance: The Blockchain is periodically maintained to remove invalid or outdated blocks to ensure the integrity of the data.

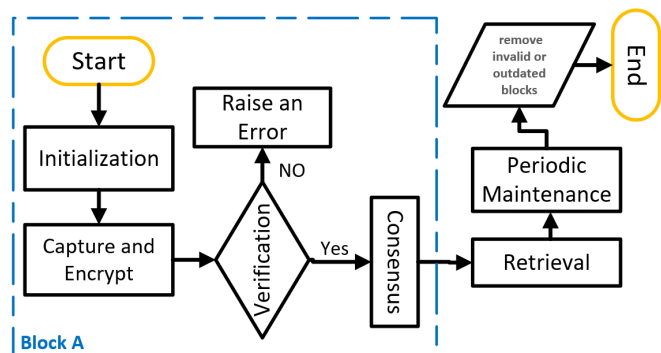


Fig. 3: Blockchain Integration Process

C. Vazka authentication

The second part of our contribution is using a novel authentication technique, called *Vazka*. By definition, authentication is a crucial process that verifies the identity of a user or system through credentials like passwords or biometric data, ensuring protection against unauthorized access and fraud. *Vazka* authentication introduces a unique and novel technique to grant access only for authorized users. It is inspired by Hash-based Message Authentication Code (HMAC) and HMAC-based One-time Password algorithm (HOTP). More precisely, *Vazka* authentication is based on data's digital signature and dynamic passwords.

In particular, the *Vazka* authentication process starts when a user requests access to stored data in smart home devices environment and consists of 2 phases: Password Generation and Password Matching. In this first phase, a dynamic password is instantly generated based on the current data storage. Then, in the next phase, an authorized user with the decryption key

In the following, we detail the *Vazka* Authentication implementation in the smart home system, as illustrated in Figure 4:

- 1) When the user request access, the main storage device should be deactivated and the data writing should be switched to a standby memory. The reason behind that is to keep stored data unchanged during the process of authentication.
- 2) Password Generation:
 - Locate stored data and hash it to get a hash value of 32 characters (128 bits)
 - Encrypt the hash value to generate a key (see Figure 5):
 - Read the hash value
 - Divide the hashvalue into 4 blocks
 - Swap blocks locations
 - Convert all blocks into binary
 - Divide the binary data into 4 new blocks
 - Swap binary block locations
 - Convert the binary data to hexadecimal, the result is a key
 - Store the key in *key.txt*
 - Convert the hexadecimal to binary (it represents the password that the user needs to get access)
 - Store the instantly generated password in *password.txt*
- 3) Password Matching:
 - User reads the public key from the *key.txt* file
 - If the user is authorized, the decryption process
 - Decrypt the key (only authorized users have the following decryption algorithm):
 - Convert the hexadecimal key to binary
 - Store the obtained binary in *result.txt*
 - Compare the content of *result.txt* with the content of *password.txt*. If there is a match, grant access to the user

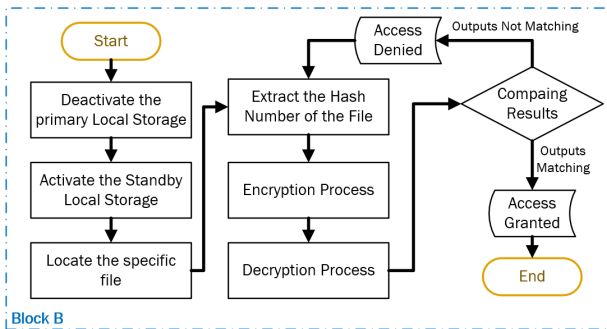


Fig. 4: *Vazka* Authentication Process

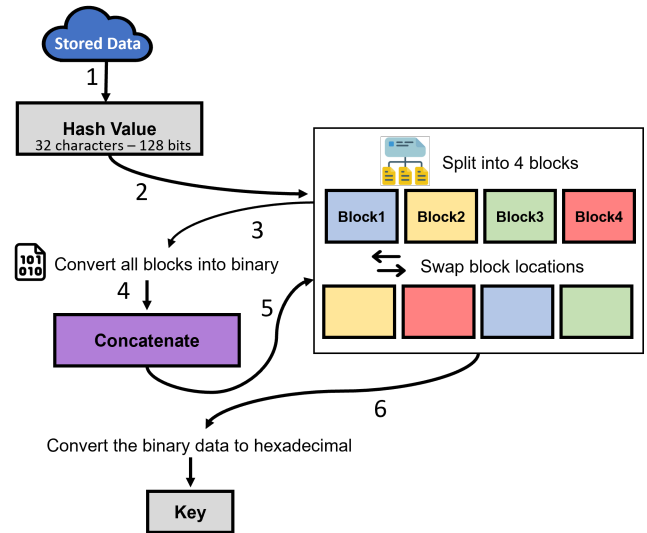


Fig. 5: Encryption Process

We made our code publicly available at <https://github.com/ksamrouth/Blockchain-Vazka-Authentication>.

The strength of our proposed *Vazka* authentication method relies in the following features:

- The key generated is not saved in a permanent storage.
- The key is generated instantly and offline with no need for the internet.
- The key is changing continuously as long as the stored content is changing.
- No need for high computational power to generate the key.
- Very difficult to be predicted because it contains combination of characters and letters.
- The password generated length is 255 characters which makes it hard to crack in short time.
- Once the access is finished, the smart home device resume the data writing on the server and hence the next access requires a different password.

IV. CASE STUDY

To illustrate the effectiveness of our proposed smart home device authentication approach, we take a case study of securing smart home security cameras with our proposed combined Blockchain with *Vazka* Authentication scheme. For testing purposes, and to evaluate our proposed solution, we consider that data collected by smart home devices are stored on a local network for the ease of testing. Our proposed solution fits this scenario as the video data captured by the cameras is 1) stored on servers, 2) continuously generated, and 3) videos do not need to be accessed on a regular basis, but only when a particular incident occurs. For instance, when a robbery occurs, and security agents need to check the video footage.

A. Case study scenario: Securing smart security camera footage

In this case study, we focus on securing the video data captured by a smart security camera. Ensuring the integrity of video data stored in the central control system is paramount. This scenario poses the following main security challenges:

- Data Integrity: Guaranteeing that video data from the camera device remains untampered throughout its transmission and storage.

- Authentication: Providing a robust, offline authentication method for the camera device that can withstand network disruptions and cyberattacks.

B. Integrating our proposed solution with the cameras system

Our solution combines blockchain technology for data integrity and "Vazka authentication" for device authentication:

- Blockchain Integration: We set up a private blockchain network within the smart camera's network, where each block stores encrypted video data from smart camera devices. The blockchain maintains a ledger of all transactions, including video data transfers between cameras and control systems. Cryptographic hashes are calculated for each data packet and recorded in the blockchain, ensuring data integrity. Any attempt to tamper with data is immediately detected, as it would require altering all subsequent blocks, which is computationally infeasible due to the blockchain's security.
- Vazka Authentication: A robbery occurred and the home owner needs to check the video footage of his smart home security camera devices. The user requests access to the stored data. At this time, the Vazka authentication process starts. First, the camera continues capturing videos but stores them on a standby storage. Then, the process of generating the dynamic instant password, described in Section III, takes place. Based on the current video footage currently stored on the server, a dynamic password is generated. Then, the user will read the key generated and decrypt it to get the password to access the video footage stored on the server. Only authorized users have the decryption algorithm, and hence only authorized users are able to deduce the password from the key. This offline authentication method ensures that even if the network is compromised, device authentication remains secure.

V. RESULTS AND DISCUSSION

A. Results of Using Blockchain

Table I shows a comparison between blockchain technology and traditional security measures in smart home systems.

With the use of blockchain, a trusted network may be established where data transfers and device interactions are recorded and encrypted, greatly lowering the possibility of unauthorized access, data breaches, and malicious activity. Hence, using blockchain in smart home devices preserves data integrity and authenticity by keeping an accurate and reliable record of events, data transfers, and device statuses by making use of the immutable and timestamped ledger provided by blockchain technology.

As a result, the obtained data is guaranteed to be genuine and legitimate, making it beneficial for analysis, diagnosis, and troubleshooting. Hence, implementing blockchain in smart home devices is an added value to these devices promoting user confidence and ensuring the security of personal data.

B. Results of Using Vazka authentication

Table II shows a comparison between Vazka authentication and existing Authentication methods. Our proposed authentication scheme adds an extra layer of security with minimum complexity in comparison with the existing authentication approaches such as multi-factor authentication approach. In contrast with traditional passwords-based authentication approaches, it supports a large number of users with minimum control overhead as passwords do not need to be stored and managed.

Finally, combining blockchain technology with the Vazka authentication method has enormous promise for boosting the security of smart home devices by maintaining data integrity and limiting unauthorized access.

TABLE I: Comparison between blockchain technology and traditional security measures in smart home systems [12] [13]

Comparison	Traditional smart home security	Smart home devices + Blockchain technology
Privacy level	Low to medium	High (ensures the privacy of each node)
Structure	Centralized	Decentralized
Scalability	Can contain large number of devices	Scalability with huge network is low
Latency	Low time	Medium to high in case of heavy traffic of data
Cost of production	Normal	High
Time for production	Normal	High
Power consumption	Low	High
Computational power	Low to medium	High
Security Level	Low to medium	High
Data integrity	High risk that data get tempered	Data integrity preserved
Complexity	Low	High
Usability	User friendly	Requires technical skills
Tamper-proof	No	Yes

VI. CONCLUSION

In this paper, we proposed to integrate Blockchain with a new authentication method to secure smart home devices. The results of this study show how to successfully handle the vulnerabilities and risks associated with smart home systems by utilizing the decentralized nature of blockchain technology and putting in place a strong authentication mechanism.

By using blockchain, we can create a secure and tamper-resistant infrastructure for smart home devices, a distributed ledger that guarantees transparency, immutability, and decentralized control. Additionally, the implementation of a new authentication method, called Vazka authentication, gives smart homes devices an extra degree of security. The system may confirm the identities of users and devices with a better degree of reliability, minimizing the risk of unauthorized access, by using an offline technique that produces

TABLE II: Comparison between Vazka authentication and existing Authentication methods

Compare	Security	User experience	Scalability	Flexibility	Cost
Username and Password	Vulnerable to password cracking	Familiar to users	Can support large number of users	Low	Low
Biometric Authentication	Strong security (depends on unique physical)	Familiar to users	Can support large number of users	Medium	Medium to high
Two-Factor Authentication	adds a layer of protection by requiring two different authentication factors	Adds extra steps but familiar to users	Can support large number of users	Medium to high	Medium
Token-based Authentication	Adds an extra layer of security	Users must have tokens	Can support large number of users	Medium to high	Medium
Vazka Authentication	Adds an extra layer of security and very hard to be predicted	This version not familiar to users but in the future many features will be added	Can support large number of users	High	Medium to high

a dynamic secret key depending on the signature of the data stored. Combining blockchain technology and the *Vazka* authentication method improves smart home device security while giving customers greater confidence and peace of mind regarding their privacy and data integrity.

VII. FUTURE WORK

Finally, it is crucial to recognize that deploying blockchain and *Vazka* authentication methods can provide certain difficulties. These include challenges with scalability, compatibility with current systems, and the requirement for widespread industry acceptance and standardization. Several future works can be done to optimize our proposed method:

- Implementing a blockchain code that reduces computing processing power to avoid delays while we have heavy traffic in data.
- Finding solutions to the increased energy consumption while using blockchain to preserve sustainability and energy efficiency.
- Creating a friendly interface allowing users to use or update the system with ease.
- Finding a solution for the internet outage problem so the blockchain works without any issues.
- Using advanced techniques to activate *Vazka* authentication: using USB, digital cards, and others.
- Combining this mechanism with other authentication techniques to be more accurate with the user identity, such as adding a fingerprint or any other solution that could be helpful to achieve this process.

REFERENCES

- [1] G. Ahmed, "Improving iot privacy, data protection and security concerns," *International Journal of Technology, Innovation and Management (IJTIM)*, vol. 1, no. 1, 2021.
- [2] M. M. R. Monjur, *Internet-of-Things (IoT) Security Threats: Attacks on Communication Interface*. ProQuest Dissertations Publishing, 2020.
- [3] Z. Zheng, S. Xie, H.-N. Dai, X. Chen, and H. Wang, "Blockchain challenges and opportunities: A survey," *International journal of web and grid services*, vol. 14, no. 4, pp. 352–375, 2018.
- [4] M. Moniruzzaman, S. Khezr, A. Yassine, and R. Benlamri, "Blockchain for smart homes: Review of current trends and research challenges," *Computers & Electrical Engineering, Elsevier*, vol. 83, 2020.
- [5] J. Park and S. Chang, "Secure device control scheme with blockchain in a smart home," *Measurement and Control*, vol. 56, no. 3-4, pp. 546–557, 2023.
- [6] T. Pecorella, L. Pierucci, and F. Nizzi, "Network sentiment framework to improve security and privacy for smart home," *Future Internet*, vol. 10, no. 12, p. 125, 2018.
- [7] A. Ghasempour, "Internet of things in smart grid: Architecture, applications, services, key technologies, and challenges," *Inventions*, vol. 4, no. 1, p. 22, 2019.
- [8] V. Kumar, N. Malik, J. Singla, N. Jhanjhi, F. Amsaad, and A. Razaque, "Light weight authentication scheme for smart home iot devices," *Cryptography*, vol. 6, p. 37, 2022.
- [9] M. Azrour, J. Mabrouki, A. Guezzaz, and Y. Farhaoui, "New enhanced authentication protocol for internet of things," *Big Data Mining and Analytics*, vol. 4, no. 1, pp. 1–9, 2021.
- [10] C. Lin, D. He, N. Kumar, X. Huang, P. Vijayakumar, and K.-K. R. Choo, "Homechain: A blockchain-based secure mutual authentication system for smart homes," *IEEE Internet of Things Journal*, vol. 7, no. 2, pp. 818–829, 2019.
- [11] T. Ahram, A. Sargolzaei, S. Sargolzaei, J. Daniels, and B. Amaba, "Blockchain technology innovations," in *IEEE Technology Engineering Management Conference (TEMSCON)*, 2017, pp. 137–141.
- [12] J. Weking, M. Mandalenakis, A. Hein, S. Hermes, M. Böhm, and H. Krcmar, "The impact of blockchain technology on business models—a taxonomy and archetypal patterns," *Electronic Markets*, vol. 30, pp. 285–305, 2020.
- [13] A. M. Gamundani, A. Phillips, and H. N. Muyingi, "An overview of potential authentication threats and attacks on internet of things (iot): A focus on smart home applications," pp. 50–57, 2018.